

Project Design Phase

Problem – Solution Fit Template

Date	10 July 2026
Team ID	
Project Name	Smart Lender – AI-Powered Loan Approval Prediction System
Maximum Marks	2 Marks

Problem-Solution Fit confirms that the solution designed for the customer actually solves the problem identified in the Ideation Phase.

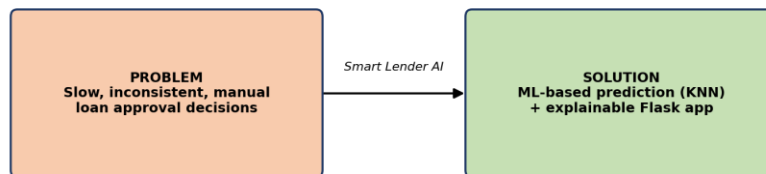


Fig 1: Problem – Solution Fit

Problem

- Applicants face long waiting times and unclear eligibility criteria when applying for a loan.
- Manual checks lead to inconsistent decisions across officers and branches.
- Lenders struggle with false approvals/rejections, which increases risk and lowers applicant trust.

Solution

- Build an ML-powered system (Decision Tree, Random Forest, KNN, XGBoost compared; KNN deployed at 84.55% accuracy) that predicts Loan_Status automatically.
- Provide a Flask web interface for applicants to submit details and instantly view the outcome.
- Translate the raw prediction into an eligibility percentage, confidence score, risk level, reasons and recommendations so the decision is explainable.
- Maintain an Applicant Profile and History page for transparency and record-keeping.

Problem – Solution Mapping

Aspect	Problem	Solution
Applicant Pain	Slow, unclear, manual approval process	Automated, near-instant ML-based prediction
Behaviour	Applicants repeatedly follow up with the branch	Applicants view result and history directly on the platform
Lender Pain	Inconsistent decisions, higher risk	Consistent model-driven decisions with explainable output